AMHARIC ACOS QUADRUPLE EXTRACTION

Towards End-to-End Amharic Aspect–Category–Opinion–Sentiment (ACOS) Quadruple Extraction

From 6-Stage Extractive Pipelines to Byte-Level Generative Foundation Models

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Target Domain Amharic Civic & Governance Commentary

Current State ByT5 Run 1 Results + Implementation & Data Audit

Source github.com/EthAIResearch/acos-amharic

v4.0 | Repository evidence update: 2 October 2026

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Amharic ACOS: Four Fields, One Exact Match

Output a set of aspect-category-sentiment-opinion tuples for each civic-domain sentence.

[aspect | category | sentiment | opinion] ; ...

NULL = an implicit aspect or opinion. None = a sentence with no quadruples.

Challenge	Evidence	Modeling consequence
Implicit components	42.71% of training quads	Need NULL prediction or implicit classifiers
Unseen opinion phrases	1,969 / 2,983 test opinions	Phrase generalization, not just vocabulary coverage
Rich morphology	Clitics and multi-word opinions	Strict boundaries remain difficult

Unseen surface phrases are not the same as tokenizer <unk> tokens. Byte coverage does not guarantee semantic accuracy.

A Civic-Domain Dataset, Not a Review Benchmark

Source text: mT5/mC4/OSCAR; Gemini annotations as described in the repository.

Original split	Sentences	Quadruples	Implicit quads
Train	39,261	41,224	42.71%
Dev	4,907	5,129	42.80%
Test	4,909	5,138	41.94%

Explicit-only training: 22,765 sentences / 23,617 quads.

Implicitness is split-specific: 42.71% in train, 41.94% in the original test set.

> LLM annotation is a provenance fact, not a quality guarantee. Human adjudication remains a research requirement.

A Cleaned Dataset Is Now Available

The cleaning pass changes both training data and evaluation annotations; old scores are historical baselines.

Clean split	Sentences	Quadruples	Implicit quads
Train	38,385	40,426	43.05%
Dev	4,892	5,110	42.80%
Test	4,908	5,133	41.89%

Merged duplicate rows: 233 train / 5 dev / 1 test. Removed leaked rows: 116 train / 10 dev.

Dropped 527 training rows over 100 tokens; resolved sentiment conflicts using train-majority labels.

Original overlap: 68 train-dev / 48 train-test / 10 dev-test texts. Clean overlap: zero.

No clean-data model metrics are recorded. Retrain and evaluate all compared models on the same data version.

22 Declared Categories; 23 Observed Labels

The taxonomy is externally fixed. Cleaning preserves labels; it does not reconcile the schema.

Observed contract

Seven civic domains; 22 labels in label_space.json and ByT5 CATEGORIES.

A 23rd label, ECONOMY#UTILITIES, occurs once in test and has no training support.

The strict ByT5 parser rejects that undeclared category.

Reporting implications

Keep the category names unchanged; reconcile the allowed list with the data owner.

Nine categories are absent from the recorded Stage 3 evaluation.

Report macro-F1 over present labels AND over all declared labels.

Explicit and NULL Fields in the Actual Dataset

Original test annotations, not model predictions or independently adjudicated reference labels.

Explicit aspect and opinion

አማርኛ መማር እፈልጋለሁ

Aspect: አማርኛ መማር

Opinion: እፈልጋለሁ

Category: PUBLIC_SERVICES#EDUCATION

Sentiment: POSITIVE | Test row: 3367

Implicit aspect; explicit opinion

አሁን ዳግም ስለመጣ ደስ ብሎኛል

Aspect: NULL

Opinion: ደስ ብሎኛል

Category: PUBLIC_SERVICES#EDUCATION

Sentiment: POSITIVE | Test row: 151

An implicit category is an annotation decision, not a recoverable span. These examples motivate human review of contextual labels.

Six Stages, Including Learned Implicit Handling

Stage 6 consumes predictions from earlier stages; gold-span component scores are diagnostics only.



1. Spans	2. Pairs	3. Category	4. Sentiment	5. Implicit	6. Assemble
Shared encoder; dual BIO heads	Union of nearest-distance pairs	Span + sentence pooling; category head	Shared PairClassifier; 3 polarities	Anchored / sentence-level classifiers	Full, explicit and implicit exact-match F1

Missing Stage 5 category/sentiment models leave a recall gap. This architecture is not inherently unable to emit NULL.

Backbone Benchmarks Across Three Subtasks

Recorded component results: Stage 3/4 use gold explicit pairs, not noisy end-to-end predictions.

Backbone	Aspect F1	Opinion F1	Category macro-F1*	Sentiment macro-F1	Sentiment accuracy
BERT-small	55.05%	39.58%	58.34%	65.27%	86.09%
AfroXLMR-base	59.93%	45.80%	60.80%	70.94%	88.57%
RoBERTa-Amharic	Not recorded	Not recorded	62.25%	71.03%	89.00%

*Present-label category macro-F1. RoBERTa: 62.25% present-label vs 36.79% all-label macro-F1.

Nine declared categories have no evaluation support. RoBERTa neutral-class F1 is only 33.78%.

Detection Is Not Complete Quadruple Recovery

Stage 5 reuses PairClassifier; absent-span masks become zero vectors while sentence context remains available.

Subtask	Accuracy	Macro-F1*	Positive-class F1
Implicit aspect / opinion anchor	80.80%	75.37%	63.81%
Implicit opinion / aspect anchor	82.50%	68.28%	47.05%
Both implicit / sentence anchor	89.71%	64.79%	35.17%
Category / implicit aspect	57.42%	51.48%	Not a detector

Nine trainable sub-models; four recorded result sets. *Category row: present-label macro-F1.

Five completion-model result sets are absent; fully implicit training keeps only the first such quad per sentence.

Error Propagation: Measured, Not an F1 Product

Issue #51 reports approximately 14% full-pipeline F1; no standalone Stage 6 metric JSON is present.

What the evidence says

Nearest-distance pairing reaches about 99.3% F1 when both input spans are gold.

Stage 1 misses and boundary shifts change the candidate pairs available downstream.

A later classifier cannot repair a missing explicit candidate in that branch.

What to report

Full-system P/R/F1 from predictions passed through every stage.

Separate component diagnostics, plus explicit and implicit quad subsets.

Do not multiply independent F1 scores or compare gold-pair accuracy with full-quad F1.

SDRN: Entity and Relation Feedback

The implementation evolved from dual-head approximation to unified 5-way CRF tagging and word-level pooling.



AfroXLMR; optional subword-to-word mean pooling

O / B-ASP / I-ASP / B-OPN / I-OPN; constrained CRF

Row-normalized attention; ESM and RSM recurrent feedback

Bidirectional span correlation; acceptance threshold

The final word-level model tags whole words. Its remaining errors cannot be explained solely by subword fragmentation.

What Helped, and What Did Not

Recorded test metric artifacts; runs change more than one factor and are not controlled single-factor ablations.

Run / change	Aspect F1	Opinion recall	Pair ceiling	Pair F1
Explicit data + weights	58.42%	48.32%	35.88%	32.75%
Exact SDRN + CRF	60.41%	44.69%	33.04%	34.16%
Asymmetric weights	59.76%	46.90%	33.51%	33.93%
+ active Dice loss	59.78%	44.66%	33.01%	32.88%
Word-level SDRN	59.96%	44.76%	32.16%	34.07%

Dice loss did not improve pair F1 in the recorded run. Thresholds differ: CRF Run 1 uses 0.20; most others use 0.50.

Seen vs. Unseen: Token Overlap & Span Boundary Breakdown

Recall across 3 granularities: full phrase, >=1 token overlap, and character trigrams for both aspect and opinion.

Span Type & Granularity	Seen in Train	Seen Recall	Unseen in Train	Unseen Recall
Opinion: Exact full phrase	1,014 (34.0%)	59.17% (600)	1,969 (66.0%)	37.63% (741)
Opinion: >=1 token overlap	2,169 (72.7%)	45.78% (993)	814 (27.3%)	42.75% (348)
Opinion: Character trigram >=50%	1,450 (48.6%)	54.41% (789)	1,533 (51.4%)	36.01% (552)
Opinion: Multi-word phrase	213 (15.2%)	62.44% (133)	1,184 (84.8%)	31.93% (378)
Aspect: Exact full phrase	1,910 (64.0%)	66.65% (1,273)	1,073 (36.0%)	43.80% (470)
Aspect: >=1 token overlap	2,550 (85.5%)	59.73% (1,523)	433 (14.5%)	50.81% (220)
Aspect: Character trigram >=50%	2,326 (78.0%)	63.67% (1,481)	657 (22.0%)	39.88% (262)

72.7% of opinions share >=1 token with train, but exact boundary recall collapses on multi-word unseen opinions (31.93%). SDRN errors stem from token-level boundary fragmentation, not pure lexical absence.

Candidate Coverage Is Not Final Pair Recall

Denominators matter: reachable gold pairs / all gold pairs versus accepted gold pairs / reachable gold pairs.

Model	Reachable / gold	Candidate recall	Accepted / reachable
SDRN CRF Run 1	979 / 2,963	33.04%	966 / 979 = 98.67%
Span-ASTE Run 1	2,403 / 2,963	81.10%	706 / 2,403 = 29.38%
Span-ASTE Run 3	2,427 / 2,963	81.91%	1,093 / 2,427 = 45.04%

Span enumeration raises reachability; relation classification then becomes the larger bottleneck. These are run-specific observations.

results/stage_aope_sdrn/afroxlmr_crf_run1/best_metrics.json | results/stage_span_aste/afroxlmr_pooled_focal_run3/best_metrics.json

Span-ASTE: Enumerate, Prune, Then Classify

Whole-span candidates replace BIO decisions; this implementation extracts explicit pairs and sentiment triplets.



Word encoding	Span features	Top-k pruning	Pair scoring	Output
AfroXLMR subwords pooled to words	All contiguous spans up to 8 words; learned width features	Separate aspect and opinion mention scores	MLP, then biaffine scoring + cross-span context	INVALID / POS / NEG / NEU; no category head or NULL candidate

Candidate spans are bounded by encoded words to avoid truncation-related indexing failures (PR #83).

High Candidate Coverage, Low Conversion

Run 1 reaches 2,403 of 2,963 gold pairs, but accepts only 706 correct pairs at its recorded operating point.

Task	Precision	Recall	F1	True positives
Aspect extraction	71.83%	39.04%	50.59%	1145
Opinion extraction	55.90%	30.52%	39.48%	900
AOPE pairs	43.55%	23.83%	30.80%	706
ASTE triplets	38.31%	20.95%	27.09%	621

Only 29.38% of reachable gold pairs become true positives. Threshold sweeps diagnose conservatism; tune on dev only.

A Better Relation Model, With Modest Gains

Deep projections, bilinear compatibility and cross-span attention replace the concatenation-only MLP.

Recorded metric	Run 1	Run 2	Change
Aspect F1	50.59%	53.76%	+3.17 pts
Opinion F1	39.48%	42.87%	+3.39 pts
AOPE pair F1	30.80%	32.82%	+2.02 pts
ASTE triplet F1	27.09%	29.37%	+2.29 pts

34.02% is Run 2's candidate conversion, not its AOPE F1. Contextual endpoints are not a direct pool of internal words.

Full-Span Pooling and Cost-Sensitive Loss

The result artifact records weighted cross-entropy. Focal loss exists in code but was OFF for this run.

Constant-time span queries

Use half-open spans [s, e), matching the annotation convention.

P[0] = 0; P[t] = sum of x[0:t].

mean(s,e) = (P[e] - P[s]) / (e - s).

h_span = [x_s; x_(e-1); mean; width].

Dimension: 3 x 768 + 20 = 2,324.

O(n) prefix preprocessing; O(1) per span for fixed hidden width.

Recorded loss configuration

Weights [INVALID, POS, NEG, NEU] = [0.1, 1.0, 1.0, 2.0].

use_span_mean_pooling = true

use_focal_loss = false

Optional focal form in code:

-alpha * (1 - p)^gamma * log(p).

Pooling and loss weights changed together; no isolated causal ablation.

src/stage_span_aste/model.py | results/stage_span_aste/afroxlmr_pooled_focal_run3/resolved_args.json | GitHub #93

More Opinions Found, Not a Universal F1 Gain

Default metric artifacts are shown together; separate sweep outputs are reported on the next slide.

Metric	Run 1	Run 2	Run 3
Aspect F1	50.59%	53.76%	55.20%
Opinion F1	39.48%	42.87%	42.89%
Opinion recall	30.52%	35.57%	46.73%
AOPE pair F1	30.80%	32.82%	31.24%
AOPE pair recall	23.83%	27.37%	36.89%

55.20% aspect F1 is best among these Span-ASTE runs, not across the repository: SDRN CRF Run 1 records 60.41%.

results/stage_span_aste/afroxlmr_run1/best_metrics.json | results/stage_span_aste/afroxlmr_pooled_focal_run3/best_metrics.json

Dev-Tuned Results Are Not Test-Selected Peaks

Record the metric file, data version, threshold-selection split and decoding rule for every reported score.

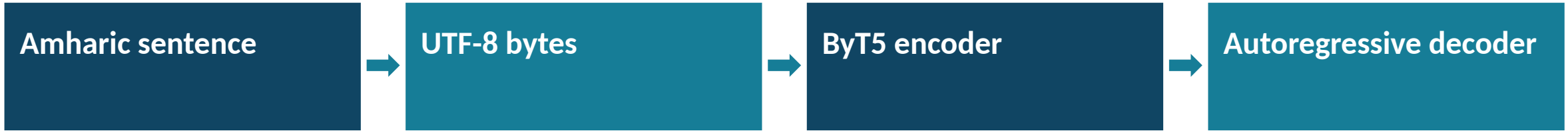
Run	Dev-selected threshold	Test F1 at dev threshold	Test-selected threshold	Peak test F1 (diagnostic)
Biaffine Run 2	0.50	32.90%	0.35	33.31%
Pooled Run 3	0.65	31.97%	0.60	32.04%

Run 3 default metrics: 1,093 pair TPs / 31.24% F1. Its separate 0.50 sweep records different values.

The often-quoted 45.78% is sweep conversion efficiency, not pair recall.

ByT5: One Model for Explicit and Implicit Quads

Byte-level encoder-decoder prediction removes the requirement that every tuple field be an input span.



Civic / governance text	256 byte values plus special tokens; no learned subword vocabulary	google/byt5-base; contextual representation	[aspect | category | sentiment | opinion]; can emit NULL

Explicit-only extractors cannot recover NULL quads as implemented. Generation is one solution; the Stage 5 classifier path is another.

The Recorded Training Setup

Resolved run arguments take precedence over current config defaults and earlier slide descriptions.

Setting	Recorded value	Why it matters
Precision / optimizer	BF16 / Adafactor	Wider exponent range; factorized optimizer state
Batch / accumulation	4 x 4 = 16 effective	Accumulation controls the effective update size
Source / target limit	768 / 256 byte-token positions	Finite input/output lengths still truncate
Epochs / learning rate	10 / 1e-4	Best checkpoint selected on dev
Dev selection / batch	First 1,000 sentences / 16	Not the full 4,907-sentence dev split

Reported engineering observations: greedy subset validation was faster than beam search; BF16 avoided the reported FP16 failure.

ByT5 Dev: Strict Quadruple and Subtask Metrics

Best checkpoint (epoch 9 per Issue #98); first 1,000 dev sentences, 1,033 gold quads.

Task	Precision	Recall	F1	TP
Full quadruple	18.20%	17.62%	17.90%	182
Explicit quadruple	15.95%	18.51%	17.14%	112
Implicit quadruple	23.49%	16.36%	19.28%	70
AOPE pair	30.00%	29.04%	29.51%	300
ASTE triplet	27.40%	26.52%	26.96%	274
ACSE (A,C,S)	30.30%	29.33%	29.81%	303
Category + sentiment	49.50%	47.92%	48.70%	495

70 implicit true positives are recorded. This is a subset-selection result, not a score on the full dev split.

ByT5 Test: Strict Quadruple and Subtask Metrics

Original test split: 4,909 sentences / 5,138 gold quads. Not a cleaned-data evaluation.

Task	Precision	Recall	F1	TP
Full quadruple	16.52%	15.78%	16.14%	811
Explicit quadruple	15.90%	18.00%	16.88%	537
Implicit quadruple	17.90%	12.71%	14.87%	274
AOPE pair	27.32%	26.10%	26.69%	1,341
ASTE triplet	24.24%	23.16%	23.69%	1,190
ACSE (A,C,S)	30.64%	29.27%	29.94%	1,504
Category + sentiment	49.40%	47.20%	48.27%	2,425

811 full-quad true positives, including 274 implicit. Raw counts and split sizes are essential context for these F1 scores.

Constrained Decoding and Linguistic Diagnostics

These features are in the codebase; no completed constrained or clitic-normalized ByT5 run is recorded.

Implemented mechanism

ByteTrie restricts category and sentiment prefixes during generation.

Evaluator exposes constrained decoding, beam width and generation penalties.

Clitic-normalized matching strips selected affixes from aspect/opinion text; category and sentiment stay exact.

Validated limitations

Category/sentiment delimiter transitions can become stuck emitting spaces.

Free-text slots still permit structural bytes; complete grammar validity is not guaranteed.

Repair and test full byte-by-byte paths before claiming zero formatting/category errors.

ByT5 Run 2: Implemented, Not Yet Measured

Canonical ordering and regularized token loss are present; curriculum training and scaling remain proposals.

Training changes

Explicit aspects sorted by a_start; implicit aspects with explicit opinions by o_start.

Fully implicit quads placed last, ordered by category and sentiment.

Label-smoothed cross-entropy: epsilon = 0.1. Planned training horizon: 15 epochs.

Experiment status

Run 2 standalone evaluation config requests 4 beams and constraints.

Training's internal evaluation does not pass the constrained option; do not assume identical decoding.

Config still points to prepared/, not prepared_clean/. No Run 2 result directory is present.

Two Evaluation Populations, Not One Leaderboard

Percent F1. The extraction rows use the explicit benchmark; ByT5 uses all original test quadruples.

Model / recorded run	Population	AOPE	ASTE	Full ACOS
SDRN CRF Run 1	Explicit pairs	34.16%	Not measured	Not measured
Span-ASTE Run 1	Explicit pairs	30.80%	27.09%	No category head
Span-ASTE Run 2	Explicit pairs	32.82%	29.37%	No category head
Span-ASTE Run 3	Explicit pairs	31.24%	28.49%	No category head
ByT5 Run 1	All quads	26.69%	23.69%	16.14%

ByT5 records 14.87% implicit-quad F1. A fair ranking requires matching data, tuple fields, implicit coverage and selection protocol.

results/stage_aope_sdrn/afroxlmr_crf_run1/best_metrics.json | results/stage_span_aste/ | results/stage_byt5_acos/byt5_base_run1/test_metrics.json

Exact Match, Subsets and Morphology Must Stay Separate

A credible comparison freezes the data version, checkpoint selection, parser and decoding settings.

What each score means

ACOS: (aspect, category, sentiment, opinion). All four must match.

AOPE: (aspect, opinion). ASTE adds sentiment. ACSE omits opinion.

Explicit: neither span is NULL. Implicit: at least one span is NULL.

ByT5 matches text tuples with multiplicity; pipeline metrics use span-index sets.

What must be controlled

Do not mix 2,963 unique explicit pairs with 2,983 explicit quads or 5,138 total quads.

Keep strict metrics primary; clitic-normalized scores are a separate diagnostic.

Report malformed-output and truncation rates; parsing can discard invalid outputs.

Use dev for thresholds. Report seeds, uncertainty and held-out sample selection.

The 500-Sample Files Are Not an LLM Leaderboard

These files were uncommitted at the audit snapshot. Their provenance limits what can be concluded.

Local artifact	Samples	Strict F1	Interpretation
llm_test_500_benchmark	500	8.02%	100 fixed predictions + 400 rule outputs
llm_benchmark_500	500	1.96%	Only 90 predicted quads; run provenance absent

The hybrid file reports 8.22% clitic-normalized F1 versus 8.02% strict F1.

This is not evidence of a ByT5 improvement, independent human agreement, or a controlled external-model comparison.

What Exists, and What Still Needs Verification

The codebase contains reusable utilities, training/evaluation entry points, configs, metric artifacts and tests.

Implemented assets

common/: span alignment, BIO decoding, label priors and shared PairClassifier.

pipeline/: assembly, inference and explicit/implicit metric views.

SDRN, Span-ASTE and ByT5 each have their own training and evaluation tools.

12 test files / 88 test functions; config and result artifacts document run settings.

Reproducibility gaps

Lightweight CI does not exercise all Torch/Transformers paths; test count is not a GPU validation claim.

Some resolved_args files omit the source config path; model checkpoints are not included.

Clean SDRN config reuses the old run output directory: choose a new one before retraining.

No clean-data or ByT5 Run 2 metric artifact is present.

src/common/ | src/pipeline/ | tests/ | .github/workflows/ci.yml | configs/stage_aope_sdrn_clean.yaml

Data and Evaluation Before Another Scale-Up

Completed implementation is not the same as a completed experiment or a validated research claim.

Priority	Next action	Evidence of completion needed
1. Data	Reconcile taxonomy; human-audit a held-out sample	Adjudicated labels and per-field agreement
2. Correctness	Repair FSM delimiter and structural-byte paths	Full-sequence generation and parser regression tests
3. Experiment	Run clean-data baselines in fresh output folders	Same splits, checkpoints and decoding protocol
4. Ablation	Isolate canonical order, smoothing and constraints	Dev-selected settings; multiple seeds / uncertainty
5. Exploration	Curriculum training or larger ByT5 + LoRA	Measured gains, resource cost and failure analysis

Already implemented: data cleaning, word-level SDRN, span pooling, canonical ordering, label smoothing, decoder scaffold and clitic diagnostics.

Repository-Wide Review Behind This Update

All 66 PRs and 39 issues were inventoried across open and closed states through #105.

Review scope

65 merged PRs; PR #9 closed unmerged. All 39 issues are closed at the snapshot.

GitHub records include 73 conversation comments, 26 reviews and 43 inline comments.

Source families: common, pipeline, Stages 1-5, SDRN, Span-ASTE and ByT5; data, configs, notebooks and tests.

Original v3.0 is retained unchanged. Source links are included in slide notes.

Evidence policy

Prefer actual metric JSON and resolved arguments over narrative summaries.

Identify conflicting artifacts instead of silently combining their numbers.

Do not promote local rule-based outputs to measured LLM performance.

A companion audit maps every issue/PR and lists corrections, additions and verification limits.

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